

As the System Architect and Full-Stack Developer, I designed and built this comprehensive ASP.NET ERP from the ground up to manage the complete lifecycle of large-scale construction projects, from initial estimation to final sale.

Architected to Solve Core Financial & Operational Challenges:

This system was engineered to give construction company owners and project managers unparalleled control and foresight.

- **Proactive Financial Health Monitoring:** The core of the system is a dynamic **"Cash Inflow vs. Outflow" projection engine**. It automatically compares projected income from sales installments against projected expenses from site activities, answering critical questions like: *"Will new sales bridge a cash gap?"* or *"Can a short-term loan be covered by future receivables?"*
- **Precision Estimation & Cost Control:** A robust estimation module handles everything from tentative budgets to final, floor-wise estimates. It intelligently manages **"Target Rate vs. Market Rate"** calculations for materials, automatically identifying cost savings (or overruns) in real-time as purchases are made.
- **Enforced Process Integrity & Workflow:** I designed and implemented a strict approval workflow for key processes:
 - **Material Indents -> Purchase Orders -> Goods Receipt -> Invoicing.** Each step is locked and logged, preventing unauthorized purchases and ensuring quantity/rate checks against estimates.
 - **Work Orders -> Site Work Done -> Running Bills.** This ensures contractors are paid only for completed, measured work at agreed-upon rates.
- **Integrated Sales & CRM:** The system manages the entire sales pipeline, from defining premises and flexible installment plans to generating automatic payment letters and tracking receipts, directly linking sales income to project milestones.

Technical Execution & My Role:

- **Role:** Sole System Architect & Full-Stack Developer (System Study, Design, Development, Deployment)
- **Technology Stack:** ASP.NET (C#), SQL Server, Crystal Reports
- **Key Features:** Configurable Voucher System, Role-Based Dashboards, Email Notifications, Automated Report Generation, Browser-Based Client for easy remote access.

Constrosoft is more than software; it's a financial and operational control center built specifically for the complexities of the construction industry.